

## MID GULF BUOY 42001, USA

WMO: 992040

Lat: 25.942N

Lon: 89.657W

Elev: 0

StdP: 14.70

Time Zone: -6.00 (W06)

Period: 94-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 57.6       | 59.8 | 42.0                          | 39.4 | 59.9 | 44.4 | 43.2 | 61.6 | 34.7                  | 67.4 | 33.1 | 67.0 | 17.9                  | 0    | 0.546 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 3.5                    | 86.4            | 77.8 | 86.2 | 77.8 | 85.7 | 77.7 | 80.4                | 83.7 | 79.8 | 83.9 | 79.3 | 84.0 | 6.3                  | 100  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|----------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)            |
| 79.5                            | 153.1 | 82.4 | 78.5 | 147.8 | 82.9 | 77.8 | 144.5 | 83.1 | 44.0          | 83.7 | 43.2 | 84.1 | 42.7 | 84.0 | 83.8           |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 31.6              | 26.8 | 24.6 | 56.2                       | 88.3 | 2.4                | 1.1 | 54.5                                               | 89.0 | 53.1       | 89.7 | 51.8       | 90.3 | 50.0       | 91.1 |  |  |
|                   |      |      | DB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |  |  |
|                   |      |      | WB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 76.8   | 69.4  | 70.0  | 71.8  | 74.7  | 78.3  | 81.7  | 83.4  | 83.7  | 82.6  | 79.2  | 74.7  | 71.2 |
|                                                                               | DBStd       | 6.09   | 4.73  | 4.42  | 4.05  | 3.05  | 2.20  | 1.71  | 1.43  | 1.49  | 1.76  | 2.87  | 3.39  | 4.60 |
|                                                                               | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD65       | 46     | 19    | 12    | 7     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 8    |
|                                                                               | CDD50       | 9768   | 600   | 560   | 676   | 742   | 878   | 951   | 1036  | 1045  | 978   | 904   | 740   | 658  |
|                                                                               | CDD65       | 4341   | 154   | 152   | 218   | 292   | 413   | 501   | 571   | 580   | 528   | 439   | 291   | 202  |
|                                                                               | CDH74       | 38427  | 247   | 238   | 460   | 1277  | 3353  | 5693  | 7145  | 7478  | 6442  | 4082  | 1421  | 591  |
|                                                                               | CDH80       | 10309  | 0     | 0     | 0     | 10    | 162   | 1496  | 2707  | 3040  | 2202  | 660   | 32    | 0    |
| Wind                                                                          | WSAvg       | 13.5   | 15.9  | 15.8  | 14.9  | 14.8  | 13.0  | 11.3  | 9.3   | 9.7   | 11.8  | 15.1  | 15.3  | 15.5 |
| Precipitation                                                                 | PrecAvg     | 41.1   | 3.5   | 2.4   | 3.1   | 3.2   | 2.1   | 3.5   | 3.3   | 4.1   | 5.2   | 4.6   | 2.8   | 3.1  |
|                                                                               | PrecMax     | 62.6   | 8.0   | 6.6   | 9.7   | 8.6   | 7.1   | 13.5  | 11.7  | 10.2  | 15.6  | 12.4  | 8.1   | 8.7  |
|                                                                               | PrecMin     | 23.2   | 0.6   | 0.2   | 0.3   | 0.1   | 0.1   | 0.2   | 0.8   | 0.6   | 1.3   | 0.5   | 0.1   | 0.5  |
|                                                                               | PrecStd     | 9.8    | 2.0   | 1.7   | 2.3   | 2.1   | 1.8   | 2.9   | 2.2   | 2.6   | 3.2   | 3.3   | 2.1   | 2.1  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 78.7  | 78.3  | 78.9  | 81.2  | 82.7  | 86.2  | 87.1  | 87.5  | 86.5  | 86.0  | 83.0  | 79.4 |
|                                                                               |             | MCWB   | 72.1  | 73.5  | 72.9  | 76.2  | 76.0  | 78.2  | 77.6  | 77.7  | 78.0  | 78.7  | 76.6  | 73.6 |
|                                                                               | 2%          | DB     | 77.5  | 77.4  | 77.9  | 80.2  | 82.2  | 85.2  | 86.3  | 86.5  | 86.1  | 84.9  | 80.7  | 78.5 |
|                                                                               |             | MCWB   | 72.4  | 73.5  | 72.7  | 75.3  | 76.0  | 78.1  | 77.5  | 77.8  | 77.9  | 78.0  | 74.1  | 72.8 |
|                                                                               | 5%          | DB     | 76.4  | 76.7  | 77.1  | 79.2  | 81.6  | 84.4  | 85.8  | 86.3  | 85.5  | 84.0  | 79.9  | 77.8 |
|                                                                               |             | MCWB   | 71.4  | 72.6  | 72.2  | 74.4  | 75.9  | 77.7  | 77.4  | 77.8  | 77.8  | 77.4  | 73.0  | 72.1 |
|                                                                               | 10%         | DB     | 75.4  | 75.6  | 76.3  | 78.2  | 81.0  | 84.0  | 85.4  | 85.9  | 85.1  | 83.1  | 79.0  | 77.0 |
|                                                                               |             | MCWB   | 70.4  | 71.4  | 71.7  | 73.4  | 75.7  | 77.4  | 77.3  | 77.8  | 77.6  | 76.5  | 72.0  | 71.3 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 75.5  | 76.3  | 76.1  | 77.5  | 81.1  | 82.0  | 80.3  | 80.8  | 80.8  | 80.4  | 77.7  | 75.7 |
|                                                                               |             | MCDB   | 77.5  | 77.1  | 76.6  | 80.2  | 81.2  | 82.9  | 84.7  | 85.4  | 84.2  | 84.3  | 82.0  | 78.6 |
|                                                                               | 2%          | WB     | 73.9  | 75.1  | 75.1  | 76.7  | 79.5  | 80.3  | 79.5  | 80.2  | 79.5  | 79.3  | 75.8  | 74.3 |
|                                                                               |             | MCDB   | 76.3  | 76.6  | 76.2  | 79.3  | 79.8  | 83.0  | 84.7  | 85.2  | 84.3  | 83.4  | 79.7  | 77.8 |
|                                                                               | 5%          | WB     | 72.9  | 74.1  | 74.3  | 76.0  | 78.4  | 79.4  | 79.0  | 79.5  | 79.0  | 78.3  | 74.3  | 73.5 |
|                                                                               |             | MCDB   | 75.7  | 76.1  | 75.9  | 78.6  | 79.8  | 83.2  | 84.5  | 85.0  | 84.2  | 82.7  | 78.7  | 77.2 |
|                                                                               | 10%         | WB     | 71.6  | 73.2  | 73.4  | 75.3  | 77.5  | 78.6  | 78.4  | 79.1  | 78.4  | 77.2  | 73.1  | 72.6 |
|                                                                               |             | MCDB   | 75.2  | 75.5  | 75.6  | 77.9  | 79.9  | 82.9  | 84.2  | 84.9  | 84.0  | 82.0  | 78.2  | 76.8 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 4.3   | 3.9   | 3.3   | 2.8   | 2.2   | 2.6   | 3.0   | 3.5   | 3.3   | 3.2   | 3.6   | 4.4  |
|                                                                               |             | MCDBR  | 4.0   | 3.3   | 2.8   | 2.4   | 2.4   | 3.0   | 3.4   | 3.6   | 3.4   | 4.2   | 4.1   | 4.2  |
|                                                                               |             | MCWBR  | 5.0   | 4.1   | 3.7   | 2.6   | 2.5   | 2.5   | 2.5   | 2.6   | 2.9   | 4.7   | 5.0   | 5.4  |
|                                                                               | 5% WB       | MCDBR  | 4.4   | 2.5   | 3.1   | 2.7   | 2.3   | 2.7   | 3.1   | 3.4   | 3.4   | 4.9   | 4.7   | 4.2  |
|                                                                               |             | MCWBR  | 5.6   | 3.4   | 4.0   | 2.9   | 2.5   | 3.0   | 2.8   | 3.1   | 3.2   | 5.0   | 6.1   | 6.1  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.369  | 0.379 | 0.389 | 0.425 | 0.438 | 0.445 | 0.454 | 0.440 | 0.426 | 0.404 | 0.382 | 0.373 |      |
|                                                                               | taud        | 2.520  | 2.497 | 2.447 | 2.342 | 2.338 | 2.394 | 2.373 | 2.432 | 2.435 | 2.472 | 2.491 | 2.516 |      |
|                                                                               | Ebn at Noon | 280    | 285   | 288   | 279   | 273   | 269   | 266   | 271   | 273   | 274   | 273   | 273   |      |
|                                                                               | Edh at Noon | 30     | 32    | 36    | 41    | 41    | 38    | 39    | 37    | 36    | 33    | 30    | 29    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1094   | 1361  | 1718  | 2029  | 2239  | 2148  | 2091  | 1959  | 1682  | 1545  | 1197  | 1033  |      |
|                                                                               | RadStd      | 98     | 147   | 164   | 166   | 82    | 126   | 84    | 96    | 121   | 103   | 125   | 82    |      |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page